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# OPTIMAL SENSOR PLACEMENT METHODOLOGY OF TRIAXIAL ACCELEROMETERS USING COMBINED METAHEURISTIC ALGORITHMS FOR STRUCTURAL HEALTH MONITORING APPLICATIONS

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## Abstract

Structural health monitoring applications present a valuable opportunity for cities to improve their infrastructures environmental life cycle, reduce operational maintenance cost and enhance their resiliency through early damage detection capabilities. The dilemma with structural health monitoring resides in balancing between the number of sensors to be installed and the amount of data that needs to be transferred, stored, and managed. These technical complications directly impact the cost efficiency of such solutions. Thus, limiting the wide range deployment of such technology. Henceforth, Optimal Sensor Placement (OSP) methodologies offers the possibility to optimize the sensor configuration that will ensure a proper and cost-efficient monitoring system to obtain the maximum modal information related to any infrastructure type.

In this study, a novel "Modal Assurance Criterion (MAC)"-based methodology with the aim of the optimal sensor placement is presented using a 410 m high rise structure. The Modal Kinetic Energy (MKE) and a MAC-based objective function are combined in the suggested methodology. The optimization problem is solved using a new hybrid metaheuristic algorithm that combines teaching-learning-based optimization (TLBO), artificial bee colonies (ABC) , and stochastic paint optimizer (SPO). With no user-defined parameters, the combination of algorithms exhibits up to 86% better fitness, avoids local optimums due to higher accuracy, and shows promising results for cost calculation optimization because it requires 50% to 70% fewer iterations compared to six evolutionary algorithms that are usually used.

## Keywords

Optimal sensor placement
Metaheuristic Algorithms
Combinatorial Optimization
Structural Health Monitoring
High rise buildings

## 1. Introduction

All major industries have benefited from the fourth industrial revolution to modernise and digitalise their production processes, except for the Architectural, Engineering and Construction (AEC) Industry which is still lagging. Doing business as usual not only has led to productivity stagnation but also contributed to reduce profits, increase risks and a damaged client relationships due to the regular time and budget overruns **[1]**. Those beforementioned problematics are the first

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cracks that expand along the lifecycle of civil infrastructures which could result in costly maintenance works **[2]** that are sometimes abandoned especially in the era of economic crisis and austerity policies. Solutions and new opportunities could be launched thanks to the new technologies that are slowly making their entry into the construction sector, thus, transforming how we design, construct, operate and maintain our buildings and infrastructure **[3]**. Digital Twins will play a key role in transforming our cities to be smart and resilient **[4]** but more importantly, increasing the infrastructures and built asset's lifecycle and keeping the operational costs at an acceptable range **[5]**.

Structural health monitoring is a Digital Twin application**[1,6]** that consists of the real time assessment of built assets structural performance for early damage detection and lifecycle longevity purposes. The benefit of such technology is presenting qualitative data on structural conditions, monitoring their behaviour, detecting anomalies and cost efficiently plan maintenance works **[7].** High rise buildings are subject to harmful environmental conditions such as ground acceleration, heavy wind forces **[8]** , or human induced solicitation like fire exposure **[9,10]** or explosions. This could compromise the structural integrity construction materials experience various physical phenomenon such as corrosion, cracks, loosen bolts and broken welds. As a result, effects of these anomalies directly affect structural behaviour in terms of vibration characteristics. Therefore, the digital twin application for structural health monitoring can use natural frequencies and mode shapes to monitor the structural integrity of civil engineering infrastructures and buildings **[11]**. Those vibration characteristics can be identified if we deploy a relevant number of sensors in optimal and suitable locations of the structure. Hence, the Optimal sensor placement (OSP) topic, has attracted a lot of interests as several studies have investigated several objective functions and assessed the relevance and reliability of different algorithms.

Since it is impossible to physically test various sensor configurations, a 3D finite element model with a set number of degrees of freedom (DOFs) is typically built for the numerical experiment. Through this model, the structure mode shapes are extracted and used as an input in the Optimal sensor placement methodology which can be formulated as a mathematical discrete optimization problem **[12]**, where the variables are combinations of sensor locations that are linked with the structures degrees of freedom.

In the literature, to assess the reliance of the sensor configuration, several objective functions have been investigated: fisher information matrix**[13]**, Information entropy **[14,15]**, Modal Kinetic Energy **[16]**, modal strain energy **[17,18]**, modal assurance criterion [11,19–21]. Once the mathematical problem is with an objective function, settled constraints and possible solutions, the Optimal Sensor Placement relevance relies on the performance of the chosen algorithm which ideally must converge into global or near global optima. Many studies have attempted to evaluate the efficiency of OSP methodology using different type of algorithms and objective functions. As an example, Chow et al [22] Yi et al [23], Downey et al [24] and Guo et al [25] used genetic algorithms and modified Genetic algorithm to solve OSP problems on different infrastructure types such as high rise buildings, transmission towers and truss bridges. The Particle swarm optimisation (PSO) Algorithm was used by Lian et al [26] and Zhang et al [27]. respectively for dams and cable stayed bridges. Other nature inspired and metaheuristic algorithms were used in several studies: Asynchronous monkey climb algorithm [28]**,** Adaptive monkey algorithm[29], Grey wolf optimizer to solve the OSP optimization problem on a steel structure [30] Firefly algorithm applied on a long span bridge [31]**.** How et al **[12]** proposed a discreate artificial bee colony approach based on modal assurance criterion for sensor placement of the canton tower. Mahjoubi et al **[11]** proposed a hypochoidal algorithm approach on the Shanghai tower.

In more recent studies, OSP techniques were employed as a starting point for structural health assessment methodologies because they were more MAC criterion-focused. In fact, to address the MAC-based objective functions, **Marco civera et al**[32]**.** suggested a damage driven OSP approach employing the non-dominated sorting genetic algorithm II. The authors' requirement for prior knowledge of the probable damage scenarios renders this work's application to high-rise

constructions inapplicable. **Zifan Zhang et al**[33]**.** evaluated various OSP strategies based on the MAC criterion. **Chaoli Zhang et al**[34]**.** investigated OSP techniques based on the FEM analysis for mode shape extraction. This was followed by a combinatorial optimization problem that aims to minimize a MAC based objective function using an evolutionary algorithm called covariance matrix adaptive evolution strategy (D-CMAES). A hybrid approach for damage identification based on mode shape modification was suggested by Subhajit Das et al[35]**.** Unfortunately, case studies involving high rise buildings were not covered by current studies.

The beforementioned studies addressed relevant issues to prove the efficiency of the Optimal Sensor Placement methodology, thus contributed to resolve several encounters related to multiple type of sensor deployment such as the uniaxial, triaxial accelerometers and strain gauge in different type of structures like dams, high rise buildings and bridges. Unfortunately, four major challenges need to be tackled. The first one, is related mainly to the efficiency of the algorithms used to solve the optimization problem whose accuracy depends on the hyper parameters that needs to be configurated for each specific OSP problem. In fact, Yi Tan et al[36]**.** detailed the many optimization techniques currently used for sensor placement, with a special emphasis on evolutionary algorithms and their improved versions since they are more effective than other approaches for combinatorial optimization. Sadly, these latter methods have a number of shortcomings, including poor global search performance, sensitivity to initial configuration randomness, and low computing speed.

The second concern is that because the OSP is a combinatorial optimization problem, it is challenging to discover an analytical solution. As a result, the chosen metaheuristic can only offer stochastic solutions, which are regrettably occasionally inaccurate. The answer lies in performing the calculation thousands of times over numerous runs, which can be expensive in terms of computation time. The third challenge resides in the interdependency between the number of mode shapes and the number of sensors to consider. Although parametric analysis is generally used to determinate the minimum number of sensors to deploy in an OSP problem, the weakness of the method lays in the considered number of mode shapes, since it directly impacts the range of the solution in the searching area. The cost of calculations for large dimensional structures is the fourth problem. According to Mahjoubi et al. [11], the precise FEM models should be widely employed for accurate OSP results because the reduced order models have been shown to be inaccurate. Unfortunately, the high dimensional FEM models can consider thousands of degrees of freedom, which raises the cost of such models' calculations significantly.

With the use of a novel Optimal Sensor Placement Methodology, the issues mentioned above are addressed in this work. The application of six metaheuristic algorithms—Firefly algorithms, Grey Wolf Optimizer, Social Group Optimizer, Artificial Bee Colony (ABC), Teaching Learning Based Optimization (TLBO), Stochastic Paint Optimizer (SPO), including the combination of SPO-TLBO-ABC—free from user-defined parameters—will be investigated using a 410 m tall tower structure as a numerical model. After choosing the optimum algorithm, low-modal-energy positions with low noise to signal ratio were first removed using a Modal Kinetic Energy and Modal Strain Energy. The idea behind this phase is to condense the solution search space while still producing equivalent or superior results, which will lower the cost of computation for a high-dimensional FEM model.

In this paper, we will present in section 2 the sensor placement methodology. In section 3, we will present the numerical prototype used for the optimal sensor placement investigations. Section 4 highlights the results and major findings while section 5 will analyse and discusses the results. Section 7 presents the conclusions.

## 2. Sensor Placement Methodology

### 2.1. Optimization Problem

The Optimal sensor placement as an optimization problem aims at identifying a limited number of locations that allows the record of as much information as possible in terms of modal and vibration characteristics of a given structure. The Modal Assurance Criterion (MAC) has been widely used

**[11,12,17,19, 21,29]** as the basis of objective functions as it describes the sensor distribution quality based on their mode shapes collinearity.

$$MAC_{i,j}(v) = \frac{\left[\varphi_i^T(v).\varphi_j(v)\right]^2}{\left[\varphi_i^T(v).\varphi_i(v)\right].\left[\varphi_j^T(v).\varphi_j(v)\right]} \quad i,j = 1,2,3 \ldots\ldots, p \tag{1}$$

$$F(v) = minf(v) \tag{2}$$

$$f(v) = \sum_{i,j=1,i\neq j} \left\{MAC_{i,j}(v)\right\}^2 \tag{3}$$

$$RMS = \sqrt{\frac{1}{p(p-1)}\sum_{j=0}^{p}\sum_{i=0}^{p} MAC_{ij}^2} \tag{4}$$

*$\varphi = (\varphi_1, \varphi_2, \ldots, \varphi_p)$ is the mode shape matrix calculated based on the finite element model; $\varphi_i$ and $\varphi_j$ is the ith and jth mode shape vector, $\varphi_i^T$ is the transpose vector of $\varphi_i$, $\varphi_i(v)$ is the ith mode shape vector, with mode shape values related the sensor location (v). and p is the considered number of vibration modes. Correlated mode shapes i and j accounts for a MAC = 1 while the opposite means a MAC = 0. The OSP problem can be formulated in a way to obtain a sensor placement that will realize a minimum MAC values between two different mode shapes i ≠ j.*

In this study, a MAC based objective function $f(v)$ was selected to evaluate the efficiency of 6 metaheuristic algorithms. Lower RMS means less correlated mode shapes, thus better results.

The optimization problem runs iteratively by selecting a potential sensor location ($v$) based on which we only keep the corresponding values in the mode shape matrix $\varphi(v)$. Following this, we use the equation (1) to determine the degree to which $\varphi_j(v)$ & $\varphi_i(v)$ are collinear; After this procedure has been performed on all of the chosen vectors of the mode shape matrix $\varphi(v)$ , the resulting values will be inserted into the equation (3). This iteration process if managed by an optimization algorithm that will iteratively select several sensor locations with the goal of acquiring a sensor layout that provides the smallest off-diagonal values of the MAC. A minimal value of the objective function, as given in equation (3) is a reflection of this fact.

Given the fact that the results are stochastic, we have executed each of the optimization process for 25 runs, with an evaluation time of 20000 iterations for each run. In this study we have fixed the mode shape number in the first 20 modes and changed the number of sensors to be deployed three times: 30 sensors, 20 sensors and then 10 sensors in order to evaluate the algorithms efficiency while investigating the dependency with different mode-sensor configurations.

### **2.2.** Search space reduction

The calculation cost is a fundamental issue when the Optimal sensor placement problems are addressed using accurate finite element models as solutions can account for millions of possibilities.

Therefore, it is important to reduce the search space of our supposed solutions before launching the optimization process of the MAC based objective function.

The modal strain energy (MSE) [17,18] and modal kinetic energy (MKE) [37,38] technics are both able to select sensor locations that have the potential for large dynamic amplitudes, which helps

enhance the signal-to-noise ratio. Hence, they present the advantage of identifying the degree of freedoms that are responsible for capturing the structure's essential dynamic features.

The MSE index can be formulated as:

$$MSE = \emptyset^T . K . \emptyset \tag{5}$$

$\phi$ as the mode shape matrix; $K$ as the structural stiffness matrix.

The calculation of the degree of freedom MSE is :

$$MSE_{ji} = \emptyset_i^T . k_j . \emptyset_{ji} \tag{6}$$

where $MSE_{i,j}$ is the $MSE$ associated with the $j^{th}$ DOF in the $i^{th}$ mode.

MKE is formulated in the following:

$$MKE = \emptyset^T . M . \emptyset \tag{7}$$

$\phi$ as the mode shape matrix; M as the Mass matrix. The calculation of the degree of freedom MKE is:

$$MKE_{ji} = \emptyset_i^T . m_j . \emptyset_{ji} \tag{8}$$

where $MKE_{i,j}$ is the $MKE$ associated with the $j^{th}$ DOF in the $i^{th}$ mode.

For this study, we kept the degrees of freedom with MSE and MKE rates that were higher than the mean value which was selected as a threshold.

### **2.3.** Optimization Algorithms

The optimal sensor placement as an optimization problem can be categorized in the combinatorial or discrete section, where solutions are permutations of element in the search space. Resolving this kind of mathematical issue is about finding an optimal solution using algorithms that can embrace three different approaches: (1) enumerative, (2) deterministic, and (3) stochastic. While the first two methods are computationally costly and sometimes unrealistic to use on a global finite element model, the stochastic algorithms are an excellent alternative because they use sophisticated steps to move from a solution to explore another one as they apply a random but guided probabilistic approach to navigate in the search space [39].

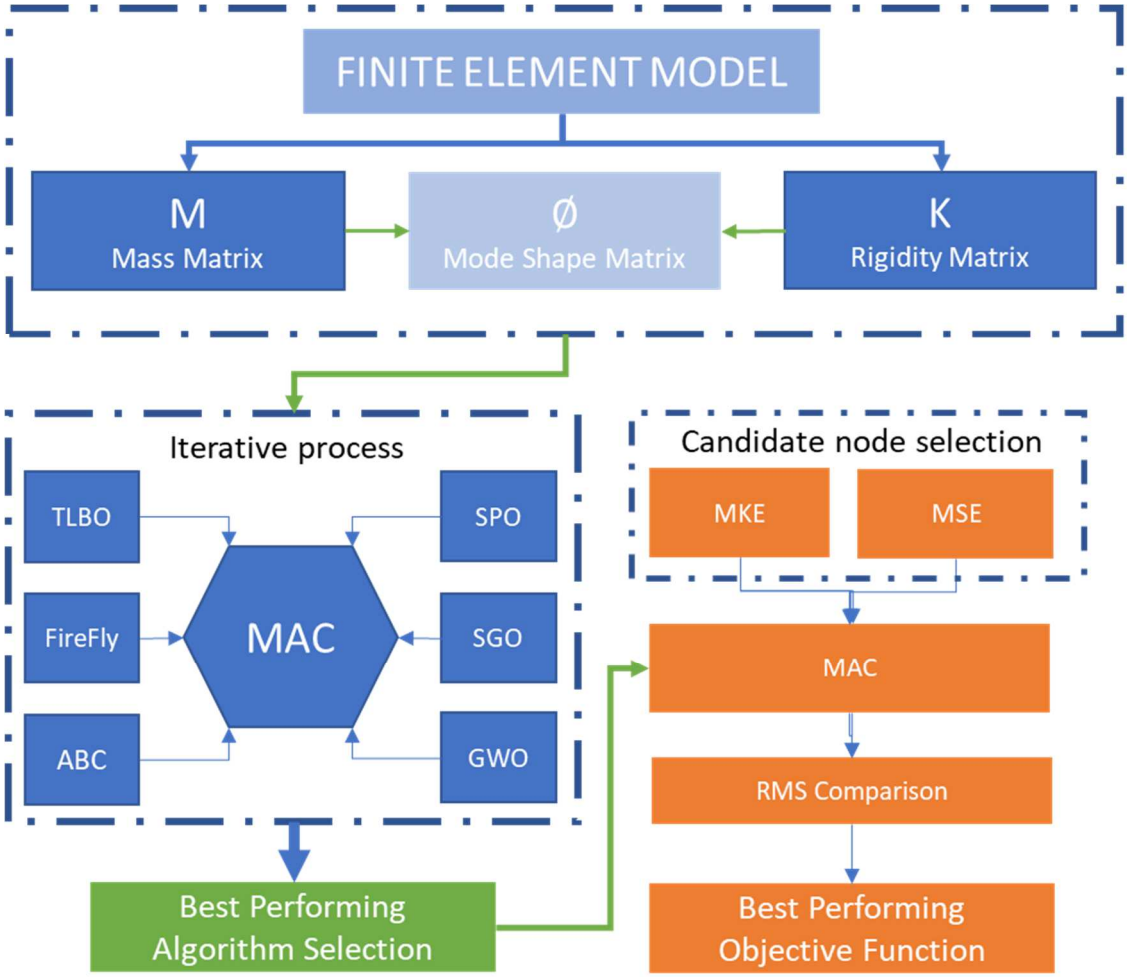
Metaheuristic algorithms are a stochastic technic that can cost efficiently solve the OSP problems. They use smart exploration methods to efficiently navigate in the search space by iteratively generating solutions to find near optimal solution. Unfortunately, they do not guarantee an exact solution as they present approximations and are usually non-deterministic. In the OSP case, where the problem is discrete and can never be deterministic, using a metaheuristic present a valuable advantage in term of cost calculation. Nonetheless, the efficiency issue needs to be addressed[40].
Once we initialize the population or search space, the metaheuristic algorithm enters a phase where it generates iteratively possible solutions. In this study we have investigated 6 metaheuristic algorithms: Firefly[40], GWO[41], SGO[42], ABC[43], TLBO [44] and the new SPO [45]**.** The focus will later be on a hybrid algorithm including (SPO), Artificial Bee Colony (ABC) and teaching-learning based optimization (TLBO) which presents the advantage of being free from user defined parameters. The purpose of this combination is to investigate its performance, its ability to avoid local optima, and evaluate the convergence speed as previous studies [46] showcased promising results compared to applying algorithms separately.

### **2.4.** Methodology

Figure 1, illustrate the methodology that we adopt in this study. We conduct first a modal analysis to extract the first 30 mode shapes of our prototype structure. Several cases will be investigated: to determinate the best performing algorithm, we fix the mode shapes that we intend to identify at the

20 first modes, then we solve the MAC based objective function for three variants, 10, 20 and 30 sensors.

In order to determinate the minimal number of sensors to be deployed, we run a parametric analysis where we fix the number of the identified mode shapes at 10, 20 then 30 and change the number of sensors.




In order to reduce the calculation cost, we shrink the solution search space by using the MKE and MSE techniques, then we solve the MAC based objective function using the best identifies algorithm, with the minimal set of sensors following the results of the previous steps.

*Figure 1 Optimal Sensor Placement methodology*

The iterative process that was adopted to evaluate the different metaheuristic algorithms is illustrated in figure 2

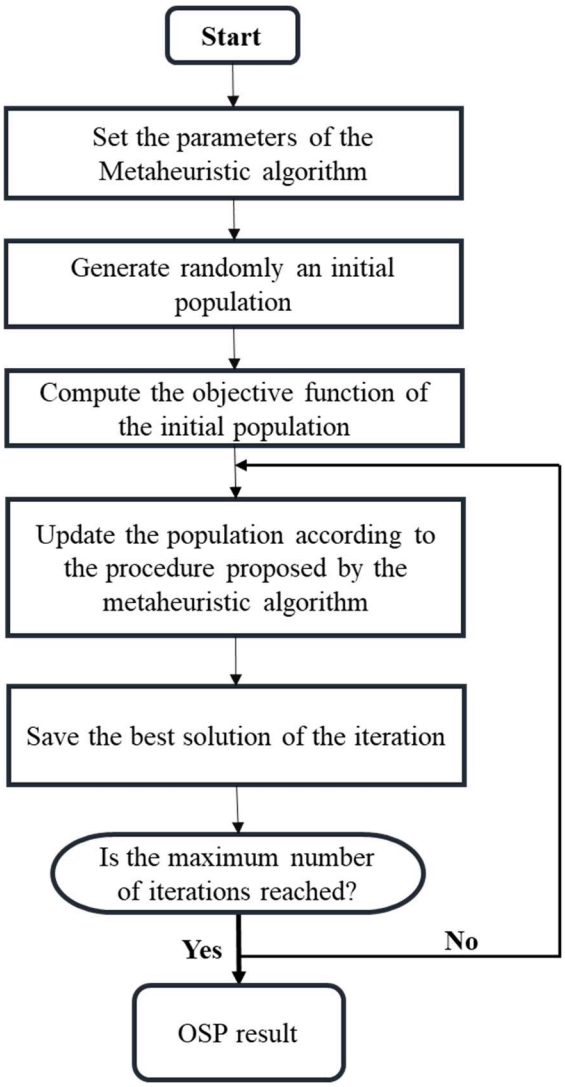



*Figure 2 Iterative process for OSP problem*

## 3. Model

In this study, we considered a similar structure to the Willis tower in Chicago that will serve as a numerical prototype to investigate the proposed optimal sensor placement methodology, using the 6 chosen algorithms to solve the MAC based objective function after reducing the solution search space by implementing the MKE and MSE techniques as was shown in the figure 1. The principal goals will be to determinate the best performing algorithm, and to identify the best technique among MSE and MKE to combine with the MAC criterion to demonstrate an effective methodology that will help reach the following sub-goals: Algorithm free from user-defined hyper-parameters, identify the minimal sensor number to deploy, results accuracy with global solutions and minimal calculation cost.

The tower is a 440 m tall building representing 110 floors. Different structural components are included to form a challenging numerical finite element model for our OSP methodology. The superstructure is composed of IPN 1000 columns, IPE 400 horizontal elements as rigid connector

beams, lateral resisting forces element as a reinforced concrete shear walls (400* 4000 cm² and floors, with tube X braced truss of D300*6. Bar elements simulates the columns, beams and the tube truss while shell elements are considered to represent the shear walls and the slabs and the hall superstructure is fixed to the ground.

The finite element model totalize 11232 steel bar elements, 1210 shell element as shear walls, 16318 nodes meaning that we have 32160 degrees of freedom to consider in our OSP investigation.

~~In this study, we have used a tower structure as a numerical prototype for high-rise building to investigate the proposed OSP methodology. It is a 440 m tall building with 110 floors. The developed full scale finite element model with beam, shell elements, shear walls and X braced frames as lateral force resisting system. The structure consists of 11232 steel beams, poles and X braced elements, 1210 shell element as shear walls, 16318 nodes and 32160 degrees of freedom.~~

~~The following table presents modal analysis characteristics of the 15 first mode shapes:~~

Table 1 presents the first 15 mode shape frequencies. Figure 4 illustrates the mode shapes for demonstration.

| Mode Number | Frequency (Hz) | Mode Number | Frequency (Hz) | Mode Number | Frequency (Hz) | Mode Number | Frequency (Hz) | Mode Number | Frequency (Hz) |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0,174 | 4 | 0,373 | 7 | 0,604 | 10 | 0,800 | 13 | 0,934 |
| 2 | 0,211 | 5 | 0,406 | 8 | 0,695 | 11 | 0,902 | 14 | 0,945 |
| 3 | 0,272 | 6 | 0,562 | 9 | 0,789 | 12 | 0,915 | 15 | 0,992 |

*Table 1 Modal analysis results*

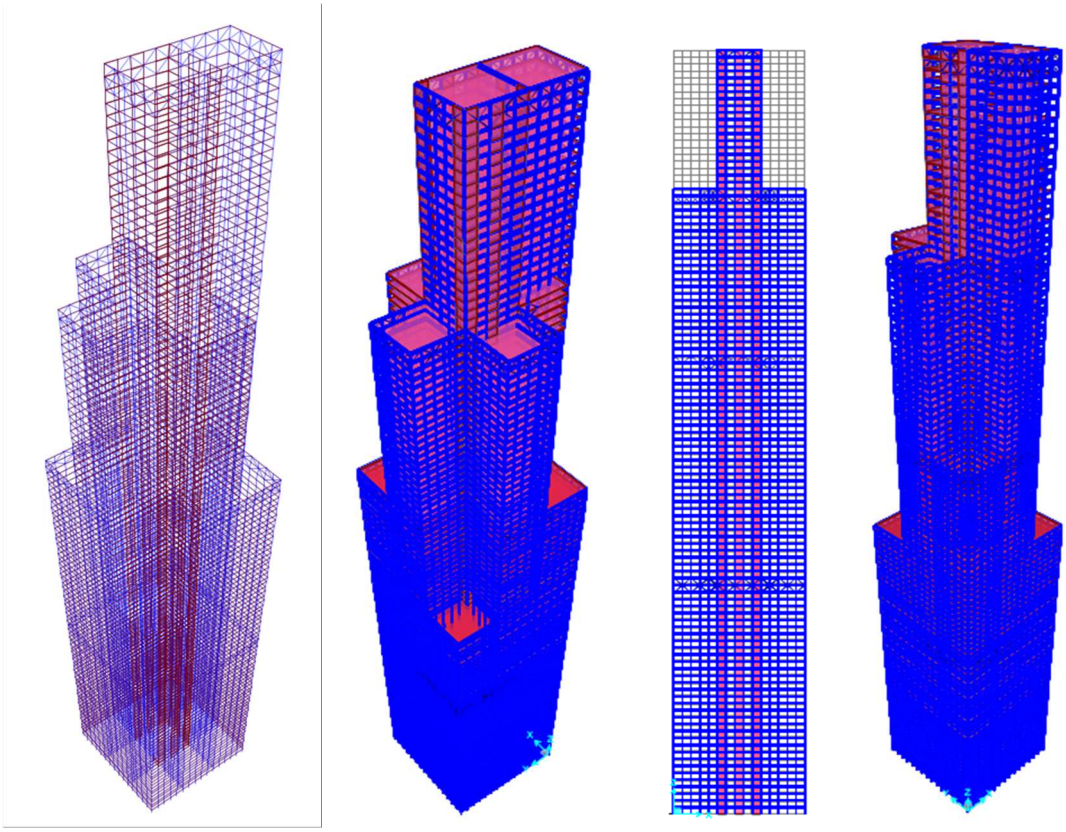


*Figure 1 structural finite element model views*

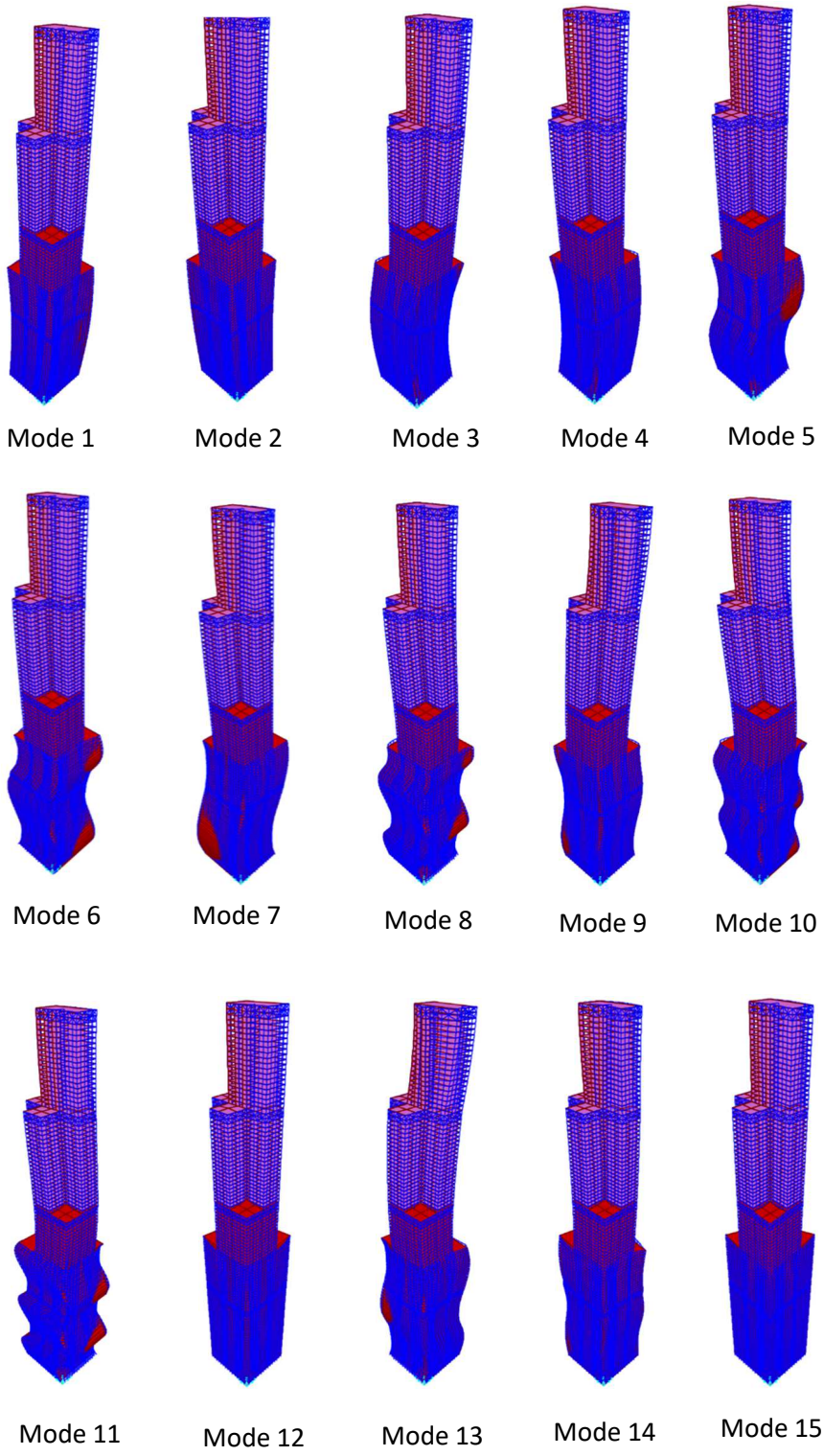



*Figure 2 The first 15 mode shapes of the Finite Element Model*

## 4. Results

### 4.1. Algorithm selection

Figures 5-a**)**, 5-b**)** and 5-c**)**, present the convergence curve of the optimization problem results using the six metaheuristic algorithms on the MAC based objective function. Three configurations were adopted: fig 5-a) 30 sensors and 20 modes, fig 5-b) 20 sensors and 20 modes and fig 5-c) 10 sensors and 20 modes.

In fact, the optimal placement of sensors is determined by finding the solution that results in the lowest possible value for the objective function which reflects the independency of the information given by the OSP solution. Beside, a low standard deviation for the numerous iterations denotes an optimization technique that can be accurate and relied upon in a consistent manner, as it shows that the sensor configuration is a global solution rather than local.

In each configuration, we had an algorithm that outperformed the others giving different results, but three algorithms have shown best performances ranking in the first three places each time: 5-a**)** 1$^{st}$ SPO, 2$^{nd}$ ABC and 3$^{rd}$ TLBO, 5-b**)** 1$^{st}$ ABC, 2$^{nd}$ SPO and 3$^{rd}$ TLBO 5-c**)** 1$^{st}$ ABC, 2$^{nd}$ TLBO and 3$^{rd}$ SPO, algorithms as they always present the minimum objective function values among the 6 metaheuristic algorithms.

Thus, we have investigated the combination of ABC-SPO-TLBO in figure 6 to conduct an OSP optimization process. Based on table 2, the combination of ABC-SPO-TLBO algorithms showed the best result in all the three configurations presented minimal “Min & Max” values of the objective functions. The power of the combination of these three metaheuristic algorithms resides in the fact that they are multi-phase algorithms, meaning that they can generate and investigate multiple solutions each iteration. The SPO has four intermediate phases, the ABC has three phases and the TLBO has two phases which all combined evaluates several sets of permutations in the same iteration. Moreover, the SPO, ABC and TLBO are algorithms free from user defined parameters which makes the result more reliant as they do not depend on the user tuning.

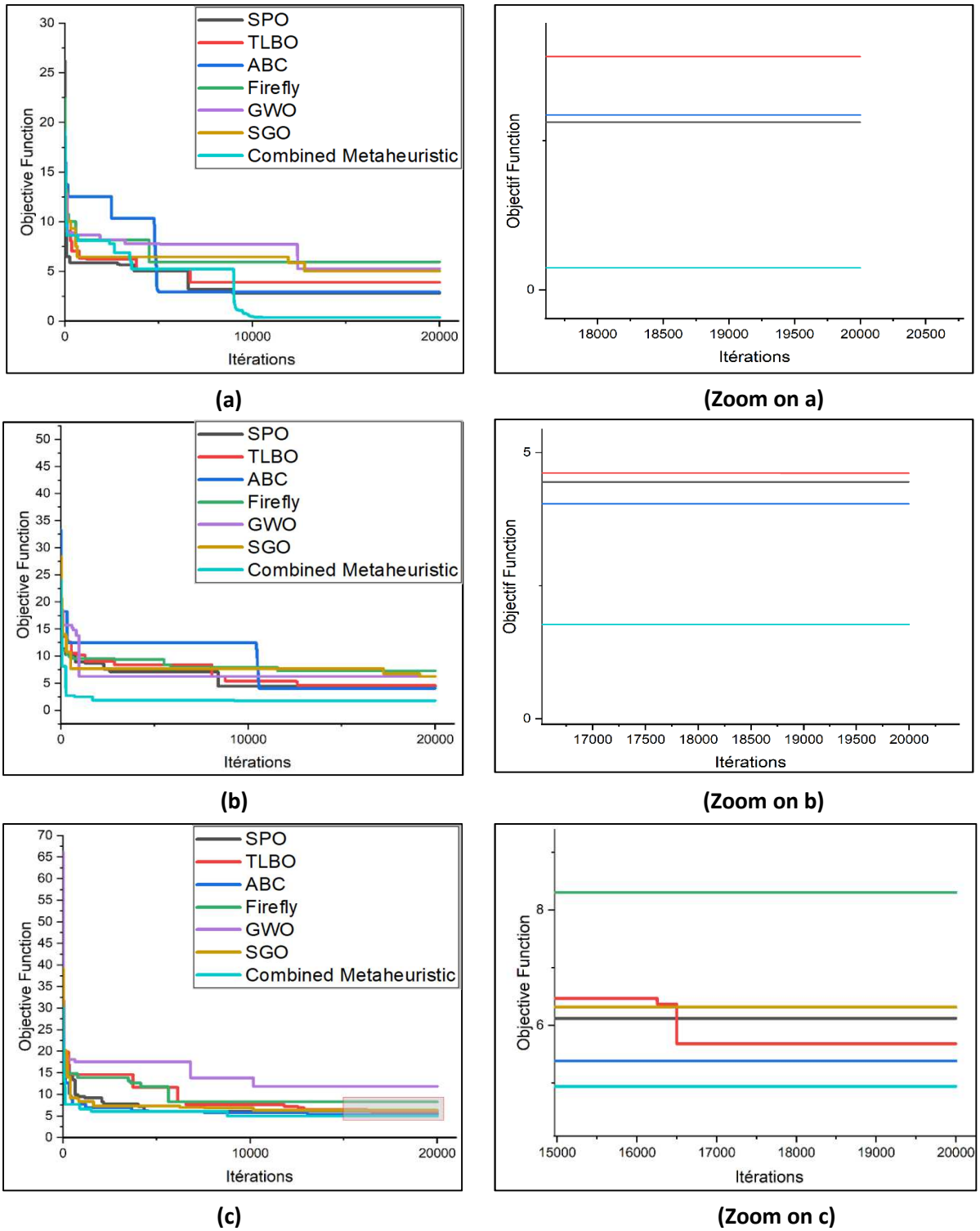



*Figure 5 Convergence curves of the OSP problem a) 30 sensors b) 20 sensors c) 10 sensors*




*Figure 3 TLBO, ABC, SPO metaheuristic combination algorithm*

## 4.2. Results uncertainties

The metaheuristic algorithms that are used to solve the OSP problem can unfortunately present some drawbacks like the local optima solutions. To limit the uncertainties, the algorithms are executed for 20000 iterations for 25 runs. Then, the accuracy of the algorithm is evaluated based on the objective function values in all runs which should demonstrate a small standard deviation. Based on the findings of the three cases :10, 20 and 30 sensors in table 2, the results indicate that the combination of ABC-SPO-TLBO has the lowest standard deviation among all the investigated algorithms which denotes an optimization technique that can be relied upon in a very consistent manner.

| | 20 Modes | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 10 Sensors | | | 20 Sensors | | | 30 Sensors | | |
| Algorithme | Min | Max | STD | Min | Max | STD | Min | Max | STD |
| ABC | 5,38E+00 | 1,24E+01 | 2,44E+00 | 4,03E+00 | 1,49E+01 | 3,77E+00 | 2,93E+00 | 1,07E+01 | 1,79E+00 |
| TLBO | 5,68E+00 | 9,84E+00 | 7,58E-01 | 4,61E+00 | 7,87E+00 | 7,18E-01 | 3,91E+00 | 5,21E+00 | 4,69E-01 |
| SPO | 5,16E+00 | 7,55E+00 | 5,52E-01 | 4,45E+00 | 7,10E+00 | 8,27E-01 | 2,81E+00 | 8,47E+00 | 1,50E+00 |
| ABC+TLBO+SPO | 4,82E+00 | 6,51E+00 | 5,17E-01 | 1,77E+00 | 3,84E+00 | 5,66E-01 | 3,68E-01 | 1,32E+00 | 3,73E-01 |

*Table 2 Min, Max and Standard deviation for MAC Values for 20 modes identification for 10 sensors, 20 sensors and 30 sensors*

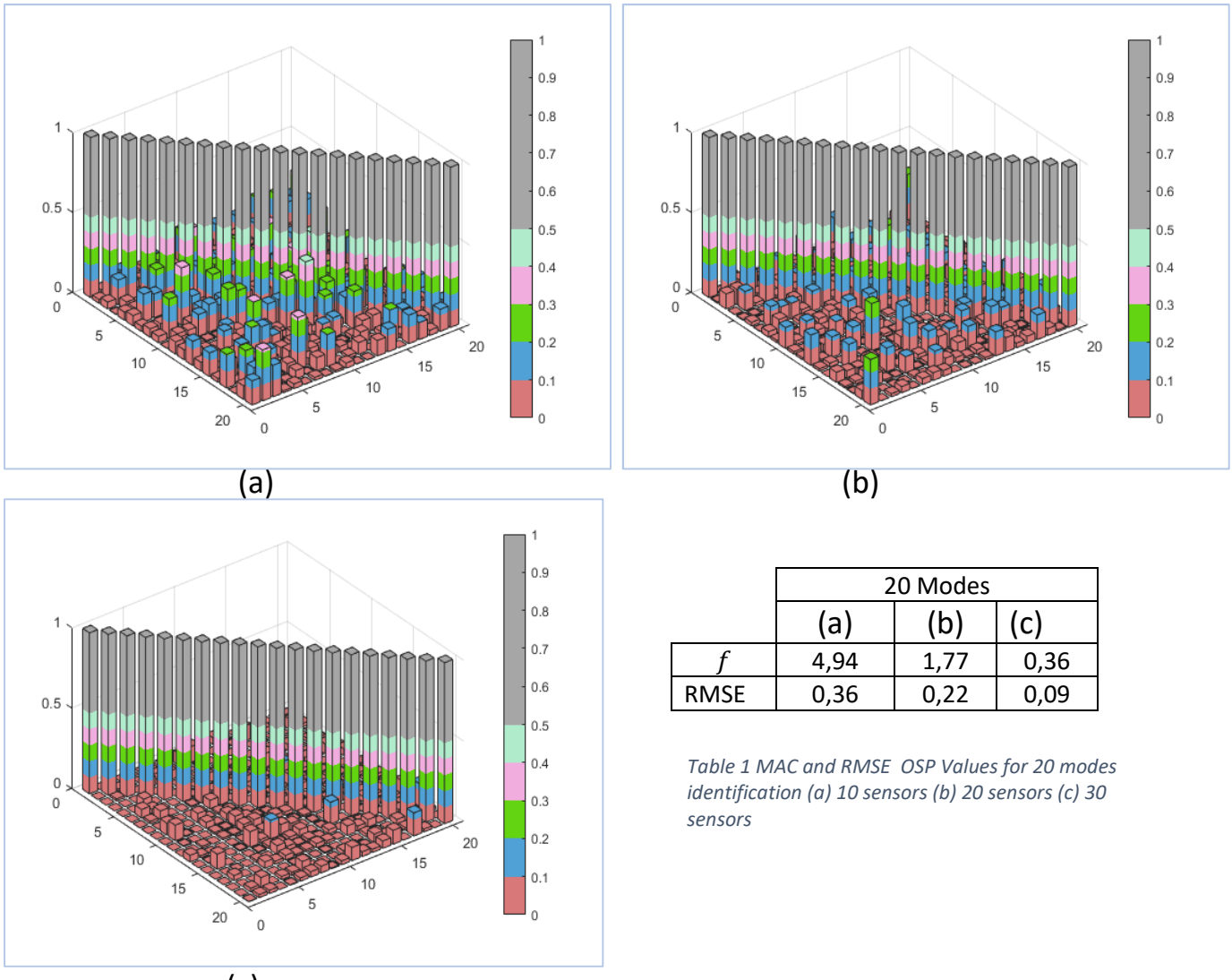



*Figure 4 Modal Assurance Criterion OSP Values for 20 modes identification (a) 10 sensors (b) 20 sensors (c) 30 sensors*

| | 20 Modes | | |
|---|---|---|---|
| | (a) | (b) | (c) |
| $f$ | 4,94 | 1,77 | 0,36 |
| RMSE | 0,36 | 0,22 | 0,09 |

*Table 1 MAC and RMSE OSP Values for 20 modes identification (a) 10 sensors (b) 20 sensors (c) 30 sensors*

### 4.3. Mode & sensors interdependency

The optimal sensor placement problem depends directly on the si
directly impacts the efficiency of the algorithms that will optimize the defined objective function.

Therefore, it is important to investigate the interdependency of sensors number to be deployed with the identified mode shape number as it constitutes the solution space and directly impact the reliance and the accuracy of the results.

In figure 8, a parametric study was performed in three different configurations: in a) we considered 10 modes, in b) 20 modes, in c) 30 modes, afterwards we executed the OSP problem using the metaheuristic combination that was demonstrated in section 4.1 as the best one. The iterations were done on 2 to 50 sensors. For each sensor, the metaheuristic combination was executed 20

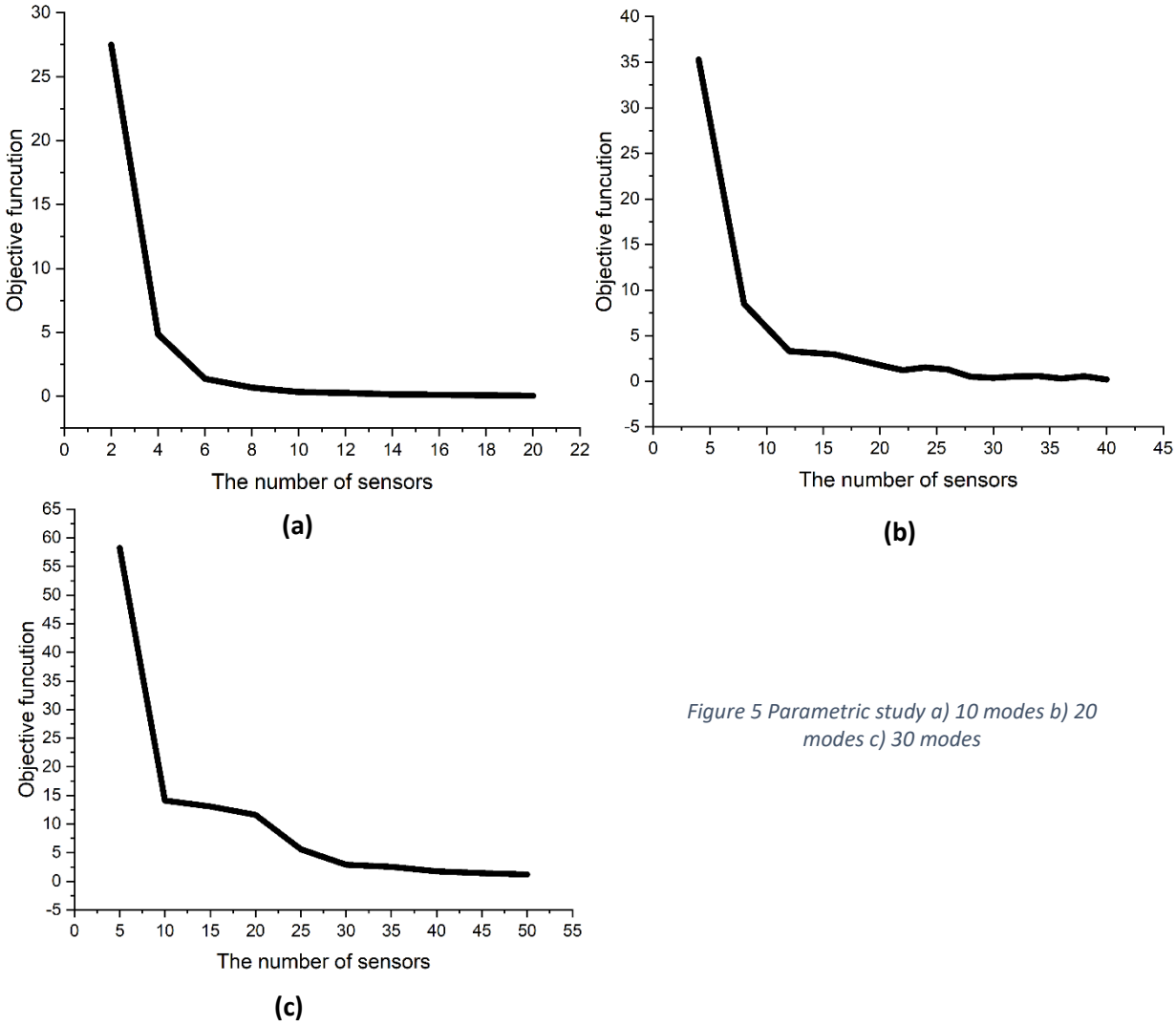



*Figure 5 Parametric study a) 10 modes b) 20 modes c) 30 modes*

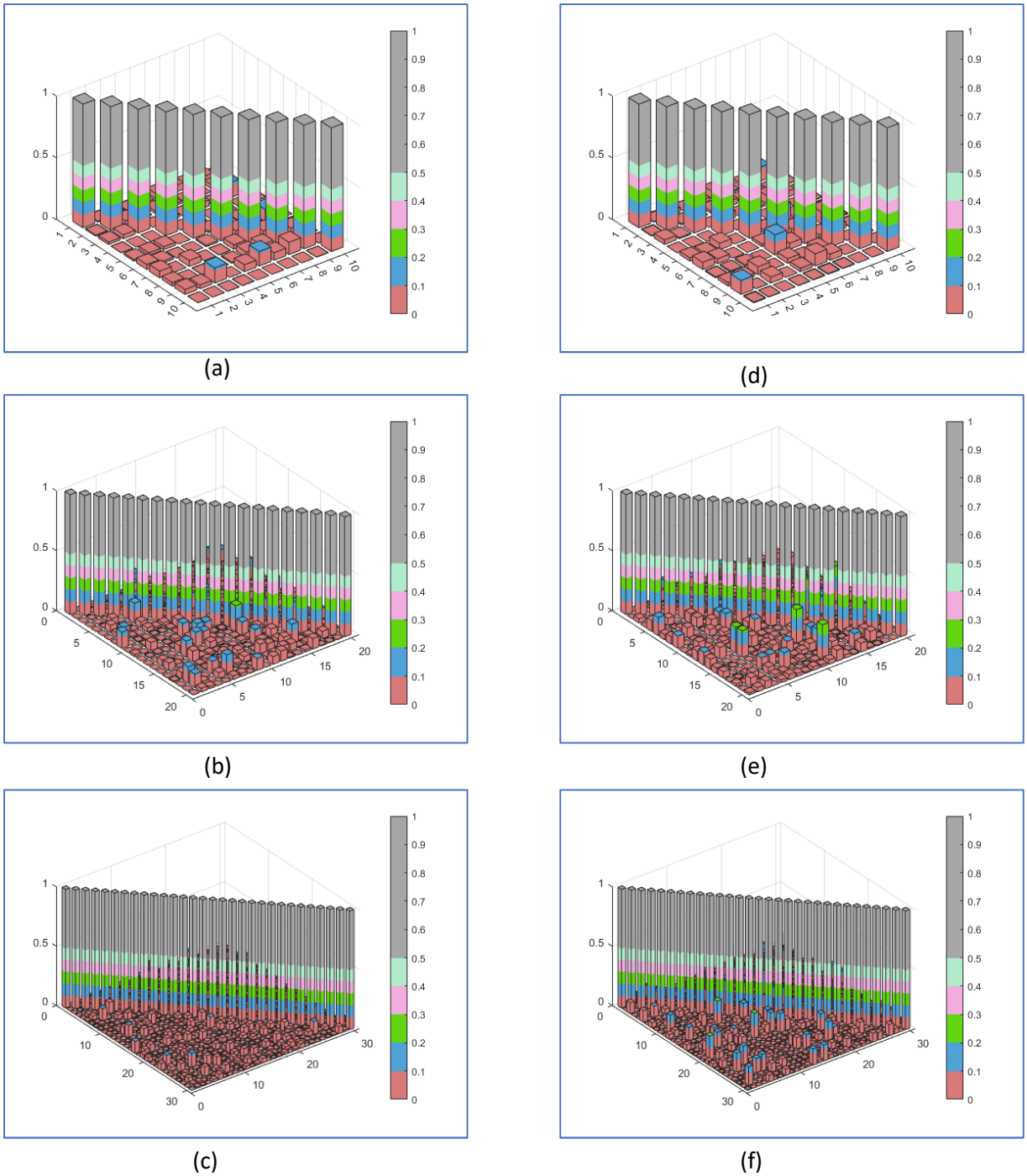
times and the minimum value of the objective function is identified. For the three configurations, the objective function values start to stabilize when the number of sensors is corresponding to the number of mode shapes to be identified.

### 4.4. Search space reduction

Performing an optimization analysis to solve the OSP problem using an accurate finite element model of a high-rise building is presenting undoubtedly high costs in calculation time. It is though very important to eliminate the sensor locations that are unlikely to be used. In this study we have investigated the combination of the Modal Kinetic Energy (MKE) and the Modal Strain Energy (MSE) before optimizing the MAC based objective function.

| | 10 Modes_10 Sensors | | | 20 Modes _20 Sensors | | | 30 Modes _30 Sensors | | |
|---|---|---|---|---|---|---|---|---|---|
| | Min | Max | STD | Min | Max | STD | Min | Max | STD |
| MAC & MSE | 1,52E-01 | 9,24E-01 | 2,58E-01 | 1,36E+00 | 3,63E+00 | 4,33E-01 | 1,76E+00 | 3,33E+00 | 4,57E-01 |
| MAC & MKE | 1,32E-01 | 6,18E-01 | 1,34E-01 | 9,95E-01 | 2,20E+00 | 4,05E-01 | 8,05E-01 | 1,80E+00 | 3,29E-01 |

446 *Table 4 MAC & MSE and MAC & MKE Values*




447 *Figure 9 Modal Assurance Criterion values combined with MKE in a) 10 sensors b) 20 sensors c) 30 sensors, and with MSE in*
448 *d) 10 sensors e) 20 sensors f) 30 sensors*

449

Figure 9 shows the results that were obtained after running the metaheuristic combination in section 4.1 in three different configurations: As recommended by the previous section, we considered the same number of sensors as the number of modes to be considered (a- 10 modes/10 sensors b-20 modes/20 sensors C-30 modes/30 sensors).

| | MKE | | | MSE | | |
|---|---|---|---|---|---|---|
| | a | b | c | d | e | f |
| $f$ | 0.13 | 0.99 | 0.80 | 0.15 | 1.35 | 1.76 |
| RMSE | 0.08 | 0.16 | 0.12 | 0.09 | 0.17 | 0.18 |

*Table 5 MAC and RMSE values combined with MKE in a) 10 sensors b) 20 sensors c) 30 sensors, and with MSE in d) 10 sensors e) 20 sensors f) 30 sensors*

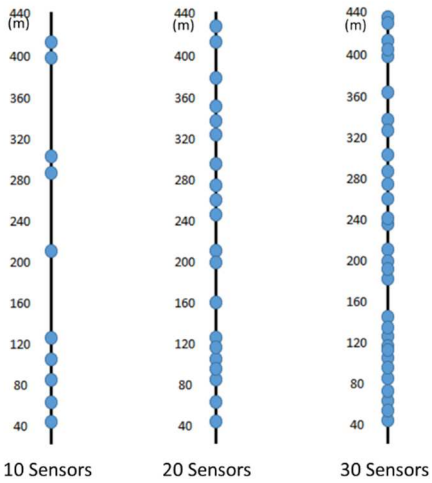



*Figure 10 OSP results with MKE-MAC based objective function using the ABC-SPO-TLBO algorithms.*

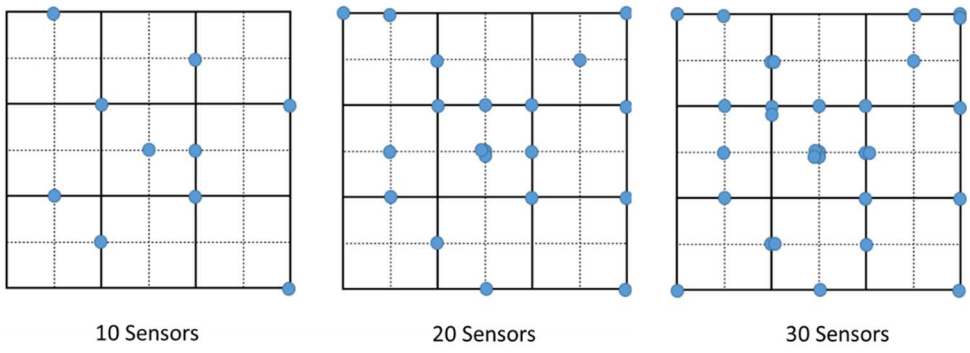



*Figure 11 plan view of the OSP results with MKE-MAC based objective function using the ABC-SPO-TLBO algorithms.*

The elevation views in figure 10 show that the accelerometers are positioned at various heights, and the plan view in figure 11 depicts their placement at various positions.

## 5. Discussion

### 5.1. Effects of the metaheuristic algorithms combination

The ABC, SPO and TLBO algorithms presented the most satisfying results for the OSP optimization problem. However, we cannot decide which algorithm is the most performing as the three achieved different results depending on the configuration that we settled for our finite element model prototype. In order to overcome this issue, we applied a hybrid metaheuristic algorithm combining the SPO, TLBO and ABC algorithms showed the most interesting results as the objective function was minimal compared to the six algorithms results. Moreover, this metaheuristic algorithm presents the advantages of being free from user defined parameters. Also, the convergence curves in figure 5 shows that the calculation cost is better as it needs less iterations to find the optimal solution. Finally, the combined metaheuristic algorithm avoids better the local optimums as shown in table 2. In fact, the SPO-TLBO- ABC combination presents the minimal “min”, “max” and “SD” values which means more accuracy and reliability compared to SPO, TLBO or ABC algorithms when considered separately.

### 5.2. Number of sensors effects on the optimization problem

Data acquisition is crucial when it comes to structural health monitoring applications. Therefore, sensor deployment with the relevant amount of triaxial accelerometers is mandatory to cost effectively gather enough information about the structure vibration. The figure 8 presents the parametric OSP solutions that was performed by incrementing the number of sensors in three different configurations: a) 10 modes, b) 20 modes c) 30 modes. The results obtained shows that the OSP solutions stabilize after the sensor number that corresponds to the considered mode shapes number.

### 5.3. Effects of the objective function hybridization

Table 4 and Figure 9 plots the results of the OSP problem using the combined metaheuristic algorithm executed on the MAC based objective function in three different methods. Combining the MAC based objective function with kinetic modal energy (MKE) and Modal Strain Energy (MSE) reduced the searching space as it only kept degrees of freedom with high modal energy. Therefore, the OSP solutions have clearly improved as it presented minimal values for the MAC based objective function, and effectively reduced the calculation cost by reducing the search space from 32160 possibility to only 4000, which is an 86 % reduction when we consider nodes with higher modal.

Table 5 shows that the MKE combination with MAC is more reliable and efficient than the MSE-MAC combination, or only MAC based objective function.

## 6. Conclusions

This paper investigated a novel combination of the modal kinetic energy with the MAC based objective function using a hybrid metaheuristic algorithm for triaxial accelerometer deployment in structural health monitoring applications in tall structures. The adopted method, which combines SPO, ABC, and TLBO, has the benefit of not requiring user-defined parameters. Furthermore, compared to six other algorithms, the hybrid metaheuristic algorithm has shown good accuracy and had an objective function that was up to 86% smaller. The local optimum dilemma is less likely to arise given the minimum standard deviation. Due to the six algorithms requiring 50% to 70% more iterations than the hybrid algorithm, calculation expenses have decreased.

The best OSP results are heavily influenced by the number of sensors that are placed. In fact, it has been established that the minimum number of sensors that should be installed in a structure is at least as many mode shapes as the user is ready to recognize. By shrinking the solution search space by more than 80%, the methodology described in this study demonstrates that the OSP problem for accurate finite element models of high-rise structures may be addressed effectively while optimizing the calculation cost. This can be done by using a cost-effective technique called the modal kinetic energy, which chooses nodes with high modal energy and is based on direct matrix multiplication of mass and mode forms. The high dimensional problem is subsequently condensed, which makes the MAC-based objective function's optimization process easier.

On the basis of finite element models, the OSP problem was previously studied. Regardless of how realistic they may seem, their modal behaviour may differ from that of a true high-rise structure. To match the digital Finite Element Model replica with the real structure, additional research should use finite element model updating techniques.

Previous studies investigated the OSP problem as a mathematical problem but omitted to assess the effect of the building type (regular, non-regular) as the non-regular structures presents torsional modes with more complicated vibration characteristics. Further studies should investigate real life buildings to evaluate more how the updated finite element models used in OSP problems will capture complicated vibrational and modal characteristics.

**Declaration of competing interest**

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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